Name: ______________________

PROPORTION,
(DIRECT PROPORTION)

Date: ________________

LO: I can solve problems involving direct proportion and find the constant of proportionality.

Direct proportion

Two quantities are in **direct proportion** if when one doubles, the other **doubles too**. The ratio between them stays constant.

Key symbol: y x means 'y is directly proportional to x'. This gives the equation y = kx where k is the **constant of proportionality**.

Quick examples

	y = kx where k = y ÷ x	constant of proportionality
	If y = 12 when x = 4, then k = 3	k = 12 ÷ 4
	5 kg costs £15, so 8 kg costs £24	k = 3 per kg
	y x means y = x	need the constant k
	If y = 10 when x = 2, k = 2	k = 10 ÷ 2 = 5

Key idea

Find k using given values, then use y = kx to find unknown values.

y = k x   k = y ÷ x

k stays constant for all pairs of values

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – finding k

y is directly proportional to x. When x = 5, y = 20. Find k and write the equation.

y = kx

20 = k × 5

k = 20 ÷ 5 = 4, so y = 4x

$\boxed{k = 4, y = 4x}$

Substitute the known values and solve for k.

Example 2 – using the equation

Using y = 4x, find y when x = 7.

y = 4x

y = 4 × 7

y = 28

$\boxed{y = 28}$

Once you have the equation, substitute any x value.

Example 3 – real-world context

5 litres of paint covers 40 m^2. How much covers 72 m^2?

k = area/litres = $40/5$ = 8 m^2 per litre

litres = area ÷ 8

72 ÷ 8 = 9 litres

$\boxed{\text{9 litres}}$

Find the rate (constant) first, then use it.

Example 4 – proportion with powers

y x^2. When x = 3, y = 36. Find y when x = 5.

y = kx^2; 36 = k × 9; k = 4

y = $4x^2$

When x = 5: y = 4 × 25 = 100

$\boxed{y = 100}$

Direct proportion can involve powers of x too.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding k

y x. Find k and write the formula y = kx.

a) x = 4, y = 12

b) x = 6, y = 30

c) x = 10, y = 45

d) x = 8, y = 56

2. Using the formula

Use your formula to find the missing value.

a) y = 3x, find y when x = 9

b) y = 5x, find x when y = 45

c) y = 4.5x, find y when x = 6

d) y = 7x, find x when y = 84

3. Real-world proportion

Solve each direct proportion problem.

a) 3 kg costs £7.50. Find cost of 8 kg.

b) 4 workers take 6 hours. How long for the same job at the same rate per worker?

c) 200 ml makes 4 pancakes. How much for 10?

d) 6 tickets cost £54. Find cost of 10.

4. Proportion with x^2

y x^2. Find the formula and missing values.

a) x = 2, y = 20. Find k.

b) Using $y = 5x^2$, find y when x = 4.

c) x = 3, y = 27. Find k.

d) Using $y = 3x^2$, find x when y = 48.

5. Mixed proportion

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) If y x, then y = kx for some constant k ☐

Correct answer: __________

Reason: _________________

b) Direct proportion always passes through (0, 0) ☐

Correct answer: __________

Reason: _________________

c) If y = 2x + 3, then y is directly proportional to x ☐

Correct answer: __________

Reason: _________________

d) k = y ÷ x when y x ☐

Correct answer: __________

Reason: _________________

e) If x doubles and y halves, they are in direct proportion ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The distance a spring stretches is **directly proportional** to the force applied. A force of **4 N stretches it 6 cm**. What force is needed to stretch it **15 cm**? What distance does **7 N** produce?

Expression: ___________________________

Simplified: ___________________________

b) y is directly proportional to **x^2**. When x = 3, y = 45. Find: (i) the value of y when x = 6, (ii) the value of x when y = 125.

Expression: ___________________________

Simplified: ___________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!